



555 TIMER LEARNING SERIES

555 Timer Monostable Alarm

Understanding one-pulse circuits through a fun burglar-alarm adventure



ENGAGE

Imagine a Burglar Alarm!

Arjun's house needs an alarm to catch the bad guys! What makes an alarm beep only once? Let's explore how a simple circuit can protect our homes and keep us safe.



Protects homes



Beeps once



Triggered by a step



Step on the mat...



...alarm beeps once!

Astable vs Monostable Modes

Comparing how a 555 Timer circuit behaves with and without repeated triggering



Astable Mode

Like a dholak drum player who never stops! It continuously blinks without needing a trigger — creating a repeating rhythm in electronic circuits.

Continuous output — no trigger needed



Monostable Mode

Like a school bell that rings just once, then goes silent. The circuit produces a single pulse when triggered, then returns to rest.

One pulse per trigger, then rest

EXPLAIN

How the Alarm Starts

Understanding the Trigger Mechanism

Pin 2 HEARS the burglar stepping on the mat — just like a hidden pressure mat under the door that activates the alarm the moment someone steps on it.



Someone steps on the mat



Pin 2 HEARS it (Trigger)

Just like a pressure mat under the door!

EXPLAIN

Resistor & Capacitor Analogies

Two everyday objects that explain how timing components work



Resistor = Water Tap

A resistor controls electrical flow just like a water tap manages water flow. Adjusting the resistance changes how much electricity passes through the circuit.

Controls the rate of flow



Capacitor = Matka (Pot)

A capacitor stores charge just like a matka stores water. The bigger the capacitor, the more charge it holds — giving a longer alarm duration.

Stores charge for longer buzzes

EXPLAIN

The Golden Rule

Understanding Alarm Timing

Bigger resistors or capacitors lead to a longer alarm buzz. This simple idea helps us understand how these components shape the performance of our burglar alarm circuit.

BIGGER

R or C



LONGER

Alarm Buzz

$$\text{Timing} = f(R, C)$$

EXPLAIN

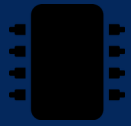
Circuit Blocks & 555 Timer Pins

Overview of Circuit Blocks



Trigger Input

(Arjun steps)



555 Timer

(Diya's chip)



Buzzer Output

(Saanvi's alarm)

Understanding the 555 Timer Pins



Pin 2 HEARS

the burglar's presence — it's the trigger input.



Pin 3 SHOUTS

the alarm — it's the output that drives the buzzer.

ELABORATE

Real-World Applications

Everyday devices that use monostable, one-pulse alarms



School Bell

A short pulse alarm for announcing class time.



Doorbell Chimes

Buzzes for 2-4 seconds to welcome visitors.



ATM Alerts

Provides instant alerts for transaction notifications.



Quiz Time!

Let's test your knowledge — Part 1 of 2

What does MONOSTABLE mean?

Two stable states

One stable state



No stable state

Multiple states

Which pin triggers the 555 Timer?

Pin 3

Pin 2



Pin 8

Pin 1



Quiz Time!

Let's test your knowledge — Part 2 of 2

What does the capacitor do in this circuit?

Stores charge



Makes sound

Turns it off

Buzzes

If we use a BIGGER resistor, the alarm will be:

Shorter

Longer



The same

Not change

Homework & Recap



Your Fun Homework

Draw the block diagram from memory and find one example of a monostable-like device at home. This will help you connect these concepts to everyday life!



Let's Recap Our Learning!

- Monostable means one pulse, then it stops.
- The trigger input starts the alarm just once.
- Resistor & capacitor control the alarm length.
- Pin 2 HEARS, Pin 3 SHOUTS!



Thank You!

We hope you enjoyed learning about the 555 Timer Monostable Alarm!



555 Timer Kit Info — Affordable for Everyone!

The 555 Timer costs ₹5–10, the buzzer ranges from ₹5–15, and you can get a full kit for under ₹50 — perfect for starting your own experiments!